Editor Guide

GAM250 / CSD2451s25

Spring AY2026

Owala

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StageBound
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Summary

This guide breaks down the main components of the Owala Editor interface to help users navigate and utilize its features for game development.

As shown below, the editor has many interface panels that will be explained in this guide.

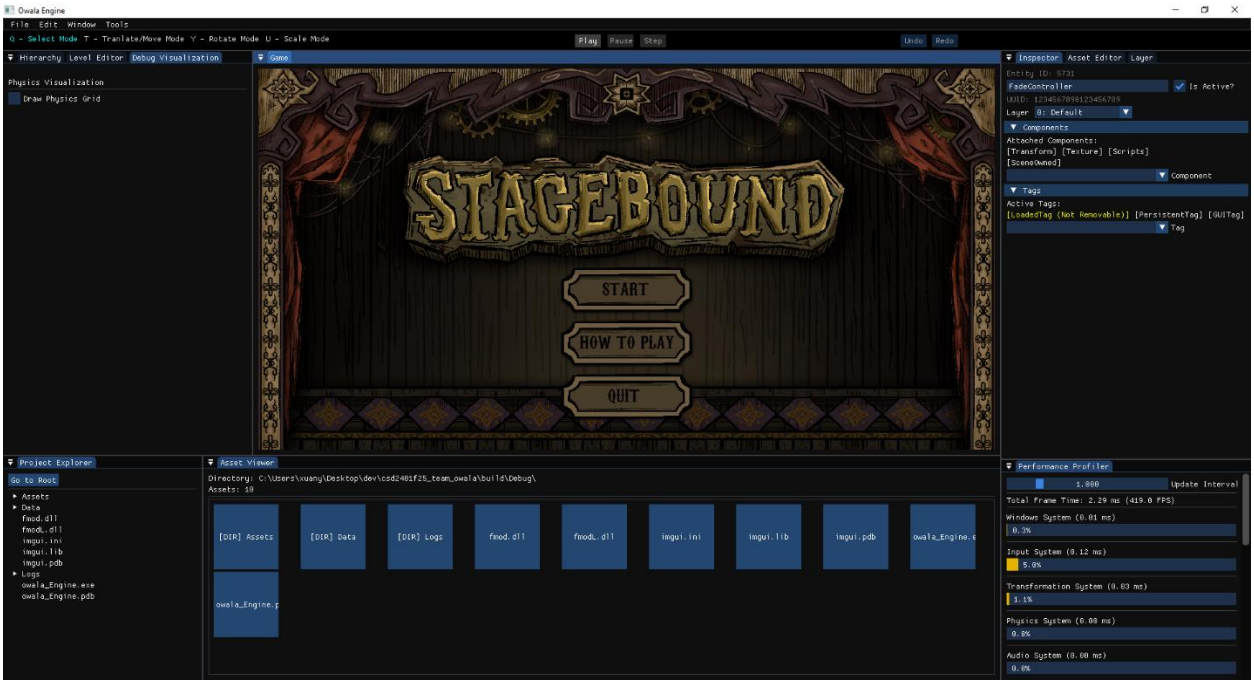


Figure 1: Owala Editor Overview

This guide will explain the functions of each panel and what information they contain, from left to right:

- **Top Left:** Hierarchy, Level Editor, and Debug Visualization panels. Above these: File, Edit, Window and Tools buttons and Gizmos keyboard shortcuts.
- **Bottom Left and Center:** Project Explorer, Asset Viewer, Tilemap Editor, Prefab Editor, Animation Editor, and FSM Enemy Editor panels.
- **Center:** Game Viewport and Toolbar.
- **Top Right:** Inspector, Asset Editor, Layer panels.
- **Bottom Right:** Performance Profiler panel.

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1.0 Project Management (Top)

The uppermost bar of the editor contains utilities required for managing and editing project settings.

1.1 File

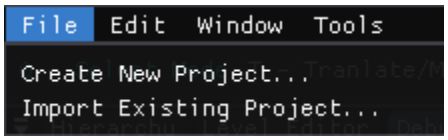


Figure 2: File Options

Create New Project: Opens a modal to generate a new game project from scratch. This tool automatically scaffolds the required folder structure (Assets, Data, Source), generates default engine configuration files (such as config.json and scenes.json), and creates a boilerplate C++ Application entry point for the game.

Import Existing Project: Opens a dialog to link an already existing game project to the editor. The tool validates the selected directory (ensuring it contains an Assets or Source folder) and automatically updates the active CMake configuration so the project can be built and run.

1.2 Edit

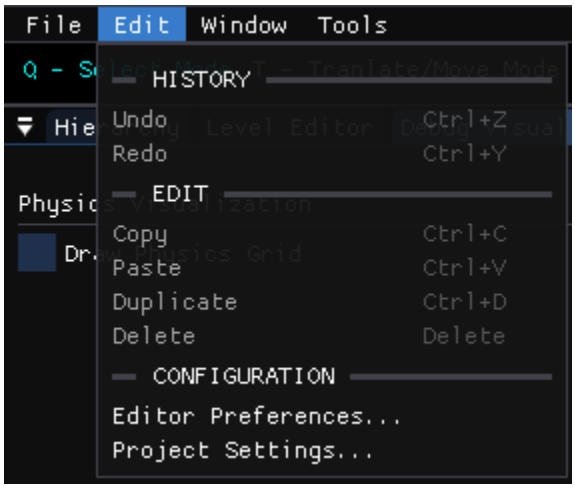


Figure 3: Edit Options

History Section

- **Undo (Ctrl+Z)**: Reverts the last modification made to the scene or an entity.

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- **Redo (Ctrl+Y)**: Reapplies the last undone operation.

Edit Section

- **Copy (Ctrl+C)**: Copies the currently selected entity and all of its component data to the clipboard.
- **Paste (Ctrl+V)**: Pastes the entity from your clipboard into the active scene as a new instance.
- **Duplicate (Ctrl+D)**: Instantly creates an exact clone of the currently selected entity within its owner scene.
- **Delete (Delete)**: Removes the currently selected entity from the world.

Configuration Section

- **Editor Preferences**: Opens a window to adjust local editor settings. Currently, this allows you to set the maximum number of history steps retained in the Undo/Redo stack.
- **Project Settings**: Opens a comprehensive configuration dashboard divided into three categories:
- **Game**: Select your *Startup Scene* and *Loading Scene*, and tag specific scenes as "Menu" or "Overlay" scenes.
- **Physics 2D**: Adjust the global World Gravity vector.
- **Performance**: Toggle the engine's FPS limit and define the maximum FPS cap.

1.3 Window

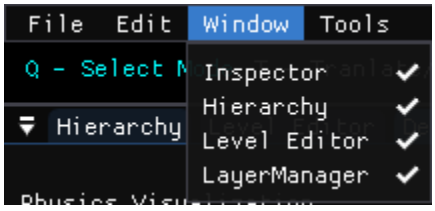


Figure 4: Window Options

This dropdown controls the visibility of your core editor docking panels. Clicking any of these will toggle the respective window on or off:

- **Inspector**: Toggles the panel used to view, add, and modify Components, Tags, and Native Scripts on your currently selected entity.

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- **Hierarchy**: Toggles the outliner panel, which displays all loaded scenes and provides a searchable list of every entity currently active in those scenes.

- **Level Editor**: Toggles the central management panel used for Scene Saving/Loading, global audio volume controls, and empty entity creation.

- **LayerManager**: Toggles the panel used to manage Render Layers (draw order) and configure the collision matrix (which physics layers interact with one another).

1.4 Tools



Figure 5: Tools Options

The Tools dropdown houses specialized sub-editors and debugging overlays.

FSM Enemy Editor: Opens a palette window that allows you to drag-and-drop Finite State Machine (FSM) enemies into the game viewport based on JSON definitions stored in the Data/FSM/ directory.

Prefab Editor: Opens a dedicated panel to modify the base components and properties of your saved Prefabs, or to delete custom prefabs entirely.

Animation Editor: Opens a toolset for configuring sprite animations, frame durations, and frame events.

Debug (Submenu):

- **Show Grid**: Toggles a visual world-space grid overlay inside the Game viewport to help with alignment.
- **Show Colliders**: Toggles the rendering of AABB (Axis-Aligned Bounding Box) physics colliders on all entities that possess them.

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2.0 Scene and Entity Management (Top Left)

This section is crucial for managing the entities within the selected scene.

2.1 Hierarchy

The hierarchy panel displays all entities present within the current scene as shown below.

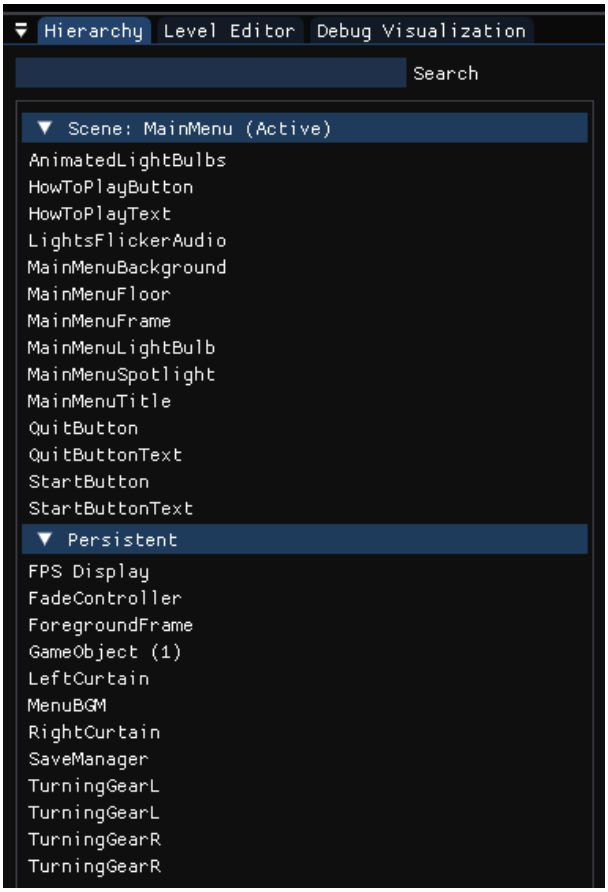


Figure 6: Hierarchy Panel

Each entity in the list can be selected by clicking on it, and it will be shown within the **Inspector** panel on the right side of the editor. There, the entity name, layer, components and tags may be viewed or edited. The selected entity will also be shown under the **World Selection** section in the Level Editor panel.

Right clicking on each entity will also bring out a context menu, where the entity can be copy and pasted, duplicated, or deleted.

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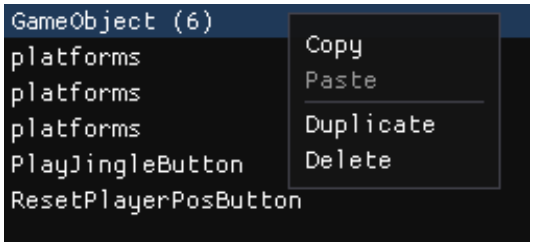


Figure 7: Entity Context Menu

2.2 Level Editor

The Level Editor panel is as shown below. Here, users can perform various Scene Management and level editing functions, such as selecting a scene under the Scenes List section to load it for editing. Upon selecting a scene, more editor scene-specific functions will be made available.



Figure 8: Level Editor Panel

The level editor consists of various functions:

- **Scene Management**
    - Displays current scene name and number of entities present in the current scene.

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    - Save button: Save current scene data to JSON file.
    - New Scene button: Create a new scene. This will bring up a modal for user to enter the scene name, which must be unique.
    - Unload Scene button: Unload the current scene. This will bring up a modal for user to confirm if they would like to save the current scene before unloading.
    - Scenes List dropdown: List of scenes currently available to load. Clicking on each scene will load that scene.
- **Entity Creation**
    - Create Empty: Create an empty entity.
    - Clone Selected Entity: Clone the currently selected entity.
- **Reset Editor Camera:** Resets editor camera.
- **Open Tilemap Editor:** Opens the Tile Palette and Prefab Palette panels on the right side of the editor, from which the user can drag and drop objects into the scene.
- **Open Prefab Editor:** Opens Prefab Editor on the right side of the editor, allowing user to edit the prefabs and save or reload its data from its JSON file.
- **Open Animation Editor:** Opens Animation Editor on the bottom middle section of the editor, allowing user to edit the animation data for any animation added to the game.
- **Global Controls**
    - Master Volume: Slider for controlling the global sound level.
    - Show Grid: Toggles visual aids for alignment and selection behavior.
- **World Selection**
    - Selected Entity: Shows entity ID of the currently selected entity. Deselect button allows user to deselect the entity.
- **Open FSM Editor**
    - Opens FSM Enemy Editor panel on the bottom middle section of the editor, from which the user can drag and drop enemy prefabs into the scene.

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3.0 Debug Visualization Tools (Top Left)

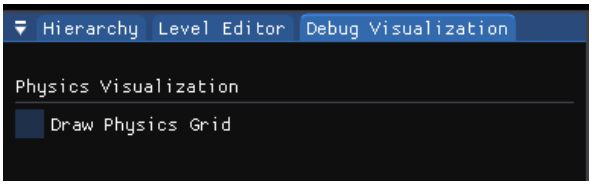


Figure 9: Debug Visualization Panel

This section provides debug visuals for the core physics engine itself.

- Draw Physics Grid: Toggles the visibility of the physics engine's broad-phase spatial grid.
    - How it works: This grid is used purely for optimization. Active cells (grid squares that contain at least one collidable object) are highlighted in green.
    - Purpose: This visualizes how the engine optimizes collision detection. By only checking for collisions between objects in the same or adjacent active cells, the system avoids millions of unnecessary checks against objects in empty space.

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4.0 Project Explorer and Asset Viewer (Bottom Left/Center)

These panels allow the user to browse project files within the editor. They are as shown below:



Figure 10: Project Explorer and Asset Viewer

The **Project Explorer** offers a hierarchical view of the project's folders and game assets, similar to a typical file directory tree.

Go to Root: Quickly navigate to the top-level project directory.

The **Asset Viewer** also allows the user to browse the project files and game assets, by clicking each folder button to navigate into the folder and the [DIR] .. button to navigate back up one level. Here, users can view image previews of the game assets.

Within the **Asset Viewer**, users may also click on a file to select it. Upon selection, it will be reflected in the **Asset Editor** panel on the top right side of the editor, where the file can be reimported or replaced. A larger image preview is also available here.

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5.0 Tilemap Editors (Bottom Center)

When the Tilemap Editor is opened by pressing the “Open Tilemap Editor” button in the Level Editor, two panels will appear on the top right of the editor: the Prefab Palette and the Tile Palette. On both panels, the Eraser button can be found at the top. Clicking on this button and then clicking on any recently placed prefab or tile will delete that entity from the scene.

5.1 Prefab Palette

The Prefab Palette allows users to level edit efficiently by using prefabs.

Users may left click on a prefab to drag and drop a prefab game object directly into the game viewport, or right click on a prefab to enter Brush Mode.

In **Brush Mode**, the prefab will be spawned at the position where the user left clicks within the game viewport.

To erase prefab objects, users can click on the **Eraser** button and proceed to click on the prefab objects to delete them.

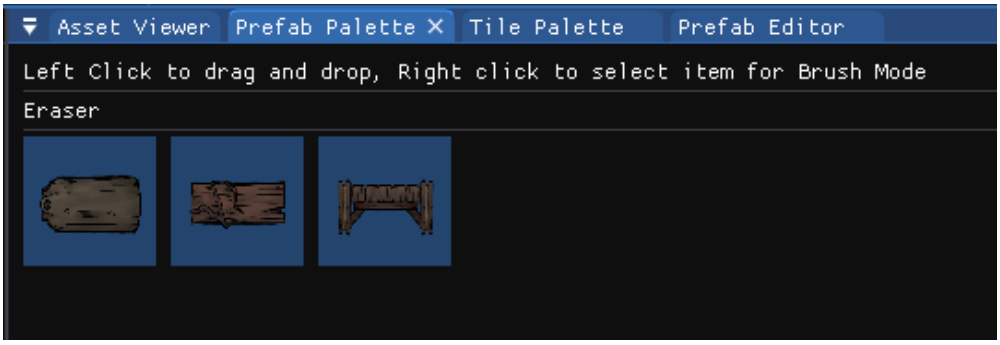


Figure 11: Prefab Palette

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5.2 Tile Palette

Similarly to the Prefab Palette, the Tile Palette allows users to build their levels efficiently within the editor by allowing them to place tiles onto the game viewport directly.

The Tile Palette works the same as the Prefab Palette. Users can left click to drag and drop a tile directly into the game viewport, or they can right click on a tile to enter Brush Mode, where they can then place the tiles wherever they left click in the game viewport.

There is also an Eraser button to allow users to delete any tiles that they placed.



Figure 12: Tile Palette

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5.3 Prefab Editor

The Prefab Editor allows users to customize their prefab game objects.

Here, they can edit the prefab properties such as the Mass, Collider size, and Texture Size. This may differ depending on the prefab being edited. Users are also able to create custom prefabs from an existing game object or delete a selected prefab.

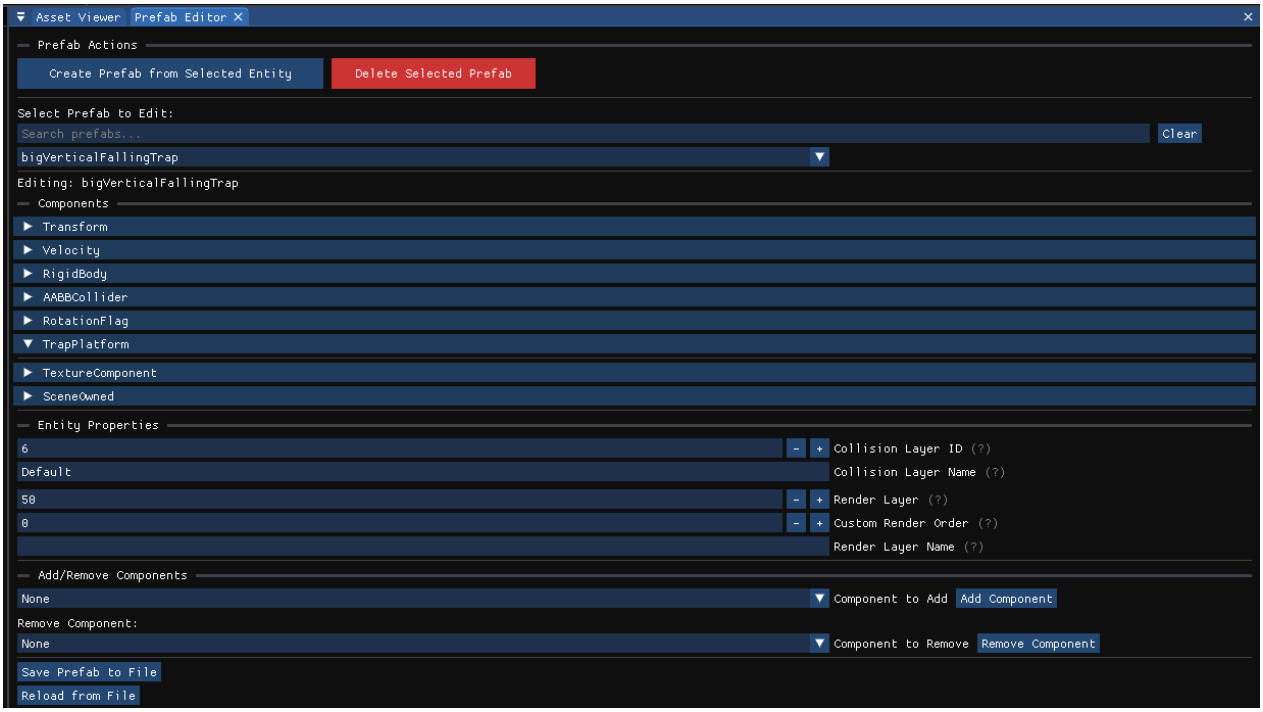


Figure 13: Prefab Editor

After users are done editing the prefab, they can click on the “Save Prefab to File” button to save their changes or they can click on “Reload from File” if they would like to discard their changes and load the original properties back.

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5.4 Animation Editor

The animation editor allows users to edit animation related data such as animation frames, animation speed and mapped sprite sheets. After editing, users may press “Save Animation” to save the new properties to disk (the Game/Data folder).

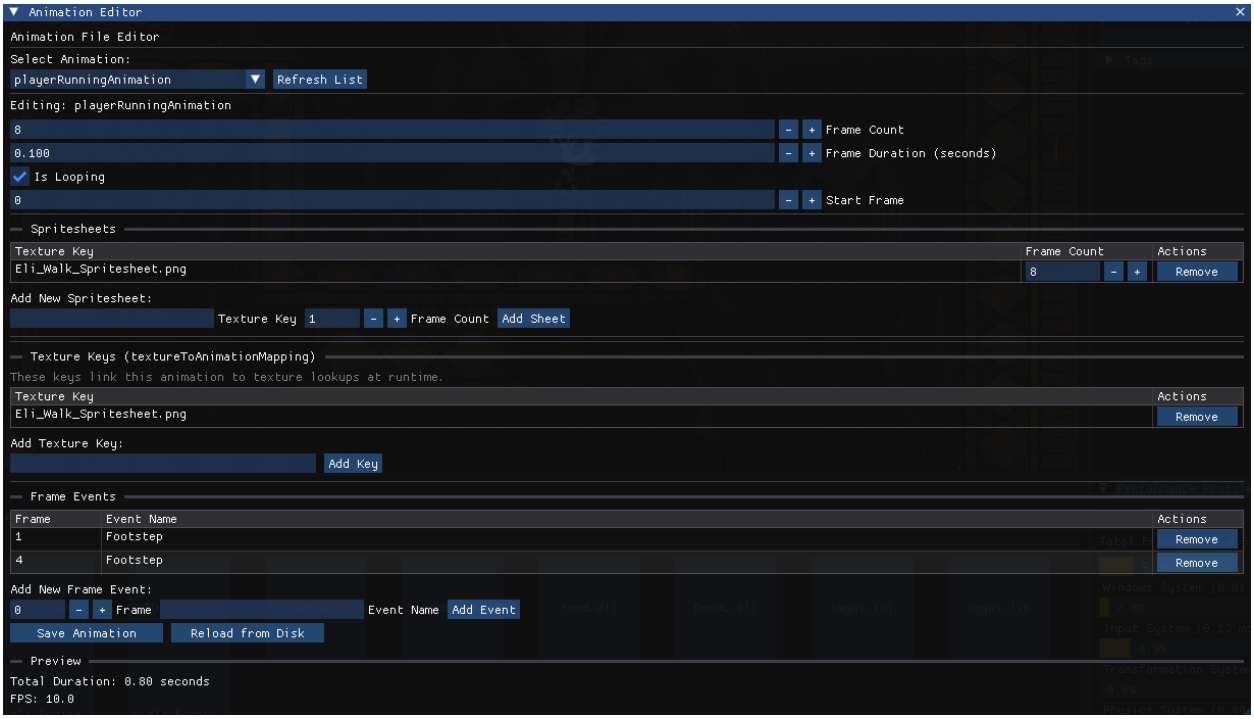


Figure 14: Animation Editor

Users may also add in their respective texture key to animation mappings, as well as add in frame events to publish based on the frame of the animation. (e.g. Footstep event to accurately track footstep audio based on frames).

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5.5 FSM Enemy Editor

The FSM Enemy Editor allows users to drag and drop a pre-configured enemy entity into the scene. It acts as a dedicated workflow for populating levels with data-driven AI without needing to manually assemble components piece by piece.

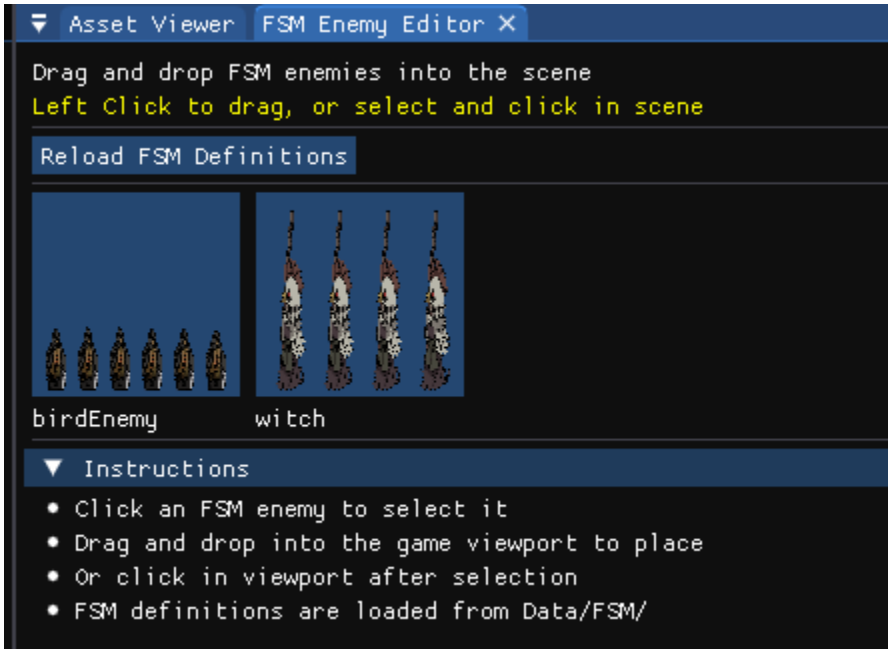


Figure 15: FSM Enemy Editor

The FSM Palette

- **Accessing the Editor:** The FSM Enemy Editor window can be opened via the top menu bar (`Tools > FSM Enemy Editor`) or by clicking the quick-access button in the Level Editor panel.
- **Automatic Population:** The editor automatically scans the `Data/FSM/` folder in your project for `.json` behavioral definitions. Any valid FSM file found is populated into the palette as a selectable item, using the preview texture defined in the JSON.
- **Live Reloading:** If a programmer adds a new FSM JSON file to the folder while the editor is running, you can click "Reload FSM Definitions" at the top of the panel to instantly refresh the palette.

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6.0 Game Viewport and Toolbar (Center)

The Toolbar and Game Viewport in the center of the Editor handle the gameplay simulation. These can be seen below in the screenshot.



Figure 16: Game Viewport and Toolbar

Within the Toolbar above the **Game Viewport**, there are:

- Gizmos Keyboard Shortcuts Information
    - The Gizmos keyboard shortcuts are displayed at the top left corner of the toolbar, with the current mode highlighted in blue to create a better user experience.
- Play/Stop Button
    - This button starts and stops the gameplay simulation. While the simulation is running, the game can be played within the game viewport, allowing the user to test out game mechanics and level design.
    - While gameplay simulation is active, the Play button also turns into a Stop Button. When the user presses Stop button, it will stop the gameplay simulation and reset all entities to their initial state.
- Pause/Step Buttons
    - The Pause button stops the gameplay simulation without resetting all the entities to their initial state. During the paused state, the user can then press the Step button to progress the physics in the game step by step, making debugging easy with this visual aid.

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- Undo/Redo Buttons
    - These buttons allow the user to undo an action they have done within the editor, as well as redo it. This feature tracks the past 30 changes (editable in Edit > Editor Preferences > Max History Stack) made and will be cleared once the scene changes.

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7.0 Inspector (Top Right)

The Inspector panel offers a more in-depth view of the currently selected entity and its associated components and tags.

As seen from the screenshot below, this panel is split into 3 sections: the top section displays the basic entity properties, the middle section displays the components, and the bottom section displays the tags.



Figure 17: Inspector Panel

7.1 Basic Entity Properties

The top section of the Inspector panel shows the basic properties of the entity, the entity's name and its collision layer. The entity name can be edited, and the collision can be adjusted using the dropdown menu.

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7.2 Components

The middle section shows the components tagged to this entity. It first displays all attached components in a list, and then a dropdown allowing the user to select a component to add, remove, or edit.

As seen below, selecting the Transform component allows for editing the position, rotation or scale in real time. Clicking on Remove Component will delete this component from the entity.

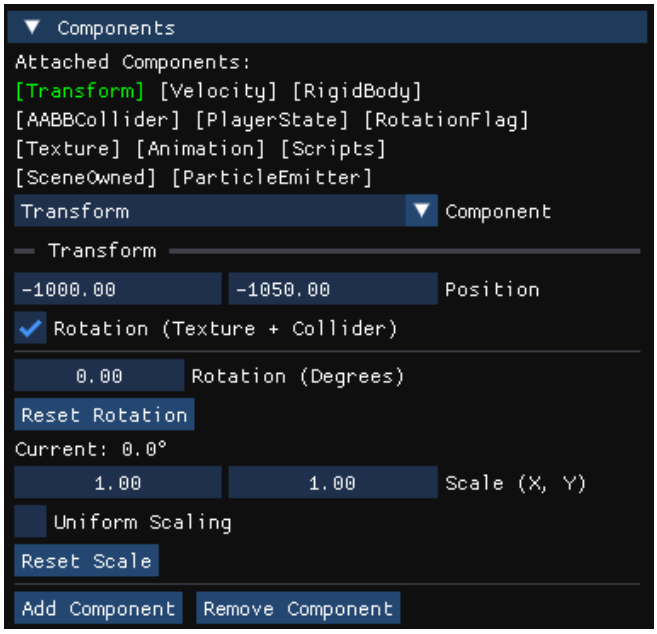


Figure 18: Selecting an attached component

As seen below, users can freely add any component to the entity.

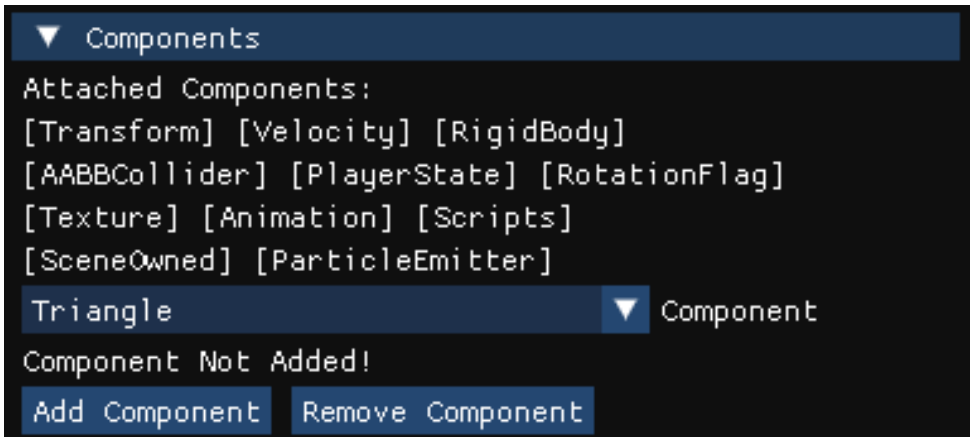


Figure 19: Selecting an unattached component

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7.2.1 Attaching Scripts to a Game Object

Here, adding the Scripts component will allow the user to attach a script to the entity.

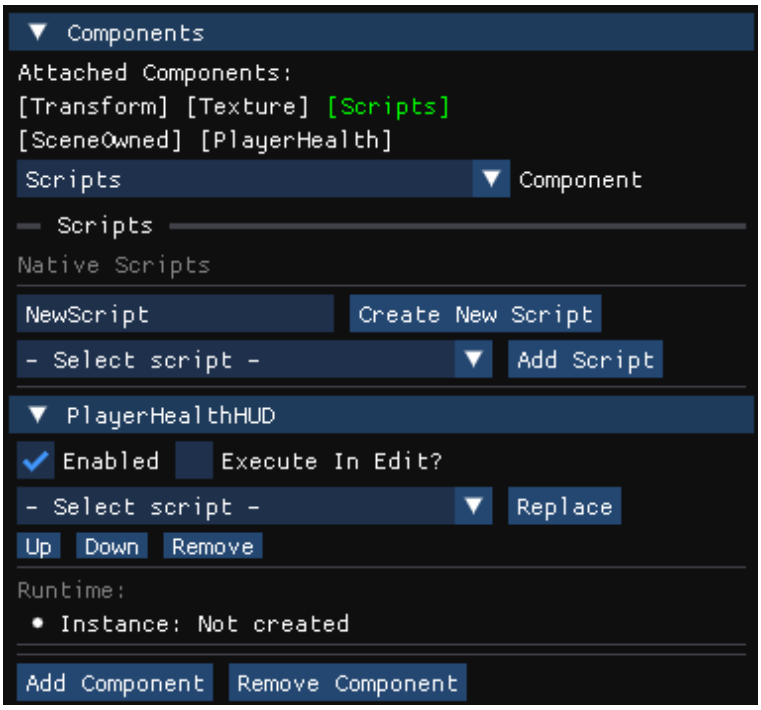


Figure 20: Attaching Scripts

With the Scripts component selected, the user can attach compiled native scripts to the entity. Once attached, the script will run their OnCreate function. Every frame, it will also run its OnUpdate function. In the future, the OnDestroy function will be implemented to run when the component is removed, or when the entity is destroyed.

The user can also choose to add a new custom script by specifying the script name beside the "Create New Script" button, then press the button to create the script. The script (both header and source file) will then be added to the Native Folder under <ProjectName>/Source/Native.

Currently, the editor does not support hot reloading for this. Hence, the user will need to run cmake and recompile again for the script to be recognized.

7.3 Tags

The bottom section displays the tags tagged to this entity. It first displays all active tags on this entity, then a dropdown for the user to add or remove tags on this entity.

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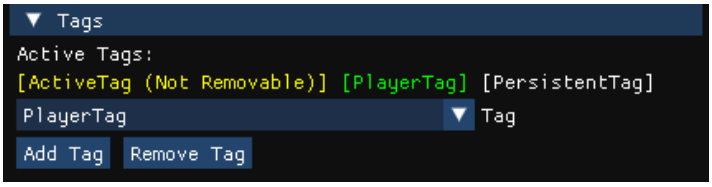


Figure 21: Selecting an attached tag

8.0 Asset Editor (Top Right)

The Asset Editor panel on the top right of the editor allows the user to see what asset they have selected from the Asset Browser, including the file path, and a larger preview of the image.

Users also have the option to reimport or replace the selected asset, as shown below.



Figure 22: Asset Editor

Reimporting a file will reload the file from the disk.

Replacing a file will allow users to drag and drop a file of the same extension into the drop zone. This preserves the old file name but updates the file with the contents of the new file. Currently, data files (json/txt), font files (ttf), and texture files (pngs/jpgs) are supported for replacement.

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9.0 Layer Editor/Viewer (Top Right)

The Layer Editor/Viewer is split into three sections:

- Scene Entity Layer Tracker
- Collision Matrix
- Entity Layer Configuration

9.1 Scene Entity Layer Tracker

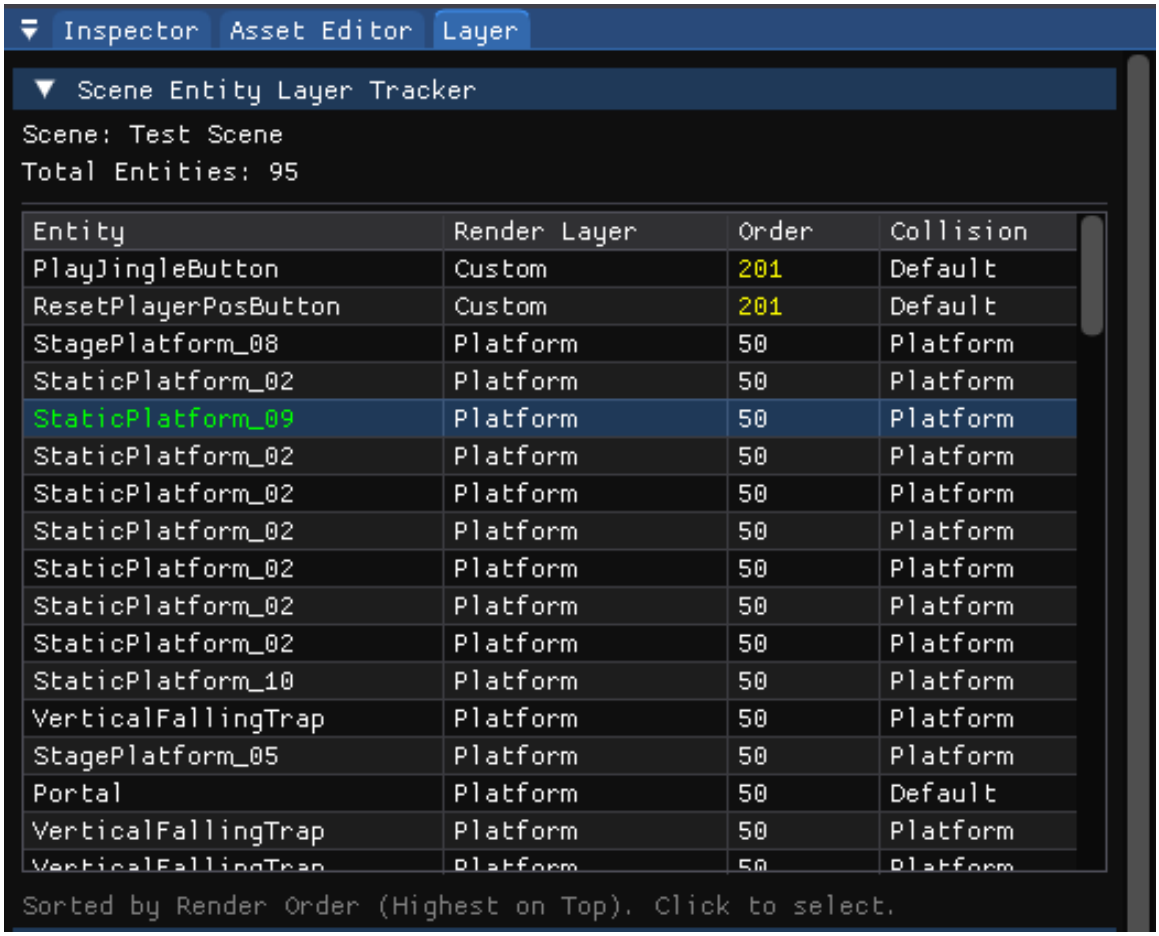


Figure 23: Scene Entity Layer Tracker

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9.2 Collision Matrix

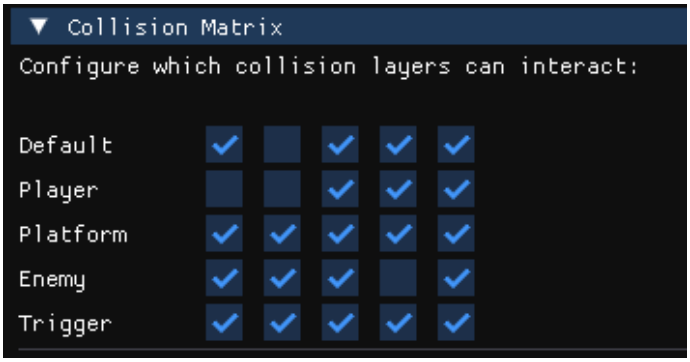


Figure 24: Collision Matrix

The collision matrix is a simplified view of collision interactions between the different layers and is easily editable.

9.3 Entity Layer Configuration

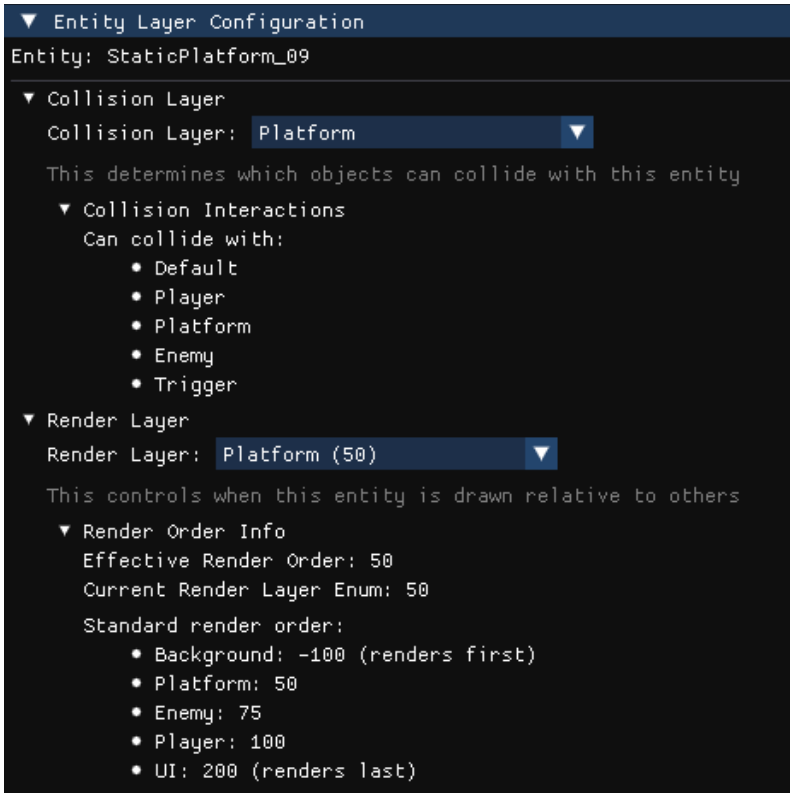


Figure 25: Entity Layer Configuration

The entity layer configuration menu allows users to change collision and rendering layers of entities. Users can also give a custom layer/value for more precise fine tuning.

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10.0 Performance Profiler (Bottom Right)

The performance profiler is a critical tool for optimization and identifying bottlenecks.



Figure 26: Performance Profiler

As seen above in the screenshot, the update intervals can be adjusted using the slider for debugging purposes. Below that, these are displayed:

- **Total Frame Time**
    - This shows the total frame time and in brackets, the frames per second.
- **System Breakdowns**
    - Here, it lists all systems and shows the processing time that each system is consuming. This is a debugging feature that allows developers to pinpoint which system is consuming the most processing time, guiding optimization efforts.

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